Final Project Proposal
Domain Adaptation in Open Language Models: Performance Analysis on Q&A and Text Generation Tasks

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Abstract

Open-source language models have demonstrated strong general-purpose performance across a wide range of natural language processing tasks. However, when applied directly to specialized domains such as legal and medical text, these models often under perform due to domain-specific vocabulary, document structure, and reasoning requirements. This limitation reduces their reliability and usefulness in high-stakes, domain-specific applications.

This project investigates the effectiveness of domain-specific supervised fine-tuning as a mechanism for adapting open-source language models to specialized domains. A pretrained open model is evaluated before and after fine-tuning on legal and medical datasets. The evaluation focuses on two representative tasks—question answering and text generation—which capture both factual understanding and generative capability within each domain.

Performance improvements are measured using standard quantitative evaluation metrics and controlled before-and-after comparisons. By analyzing results across domains and tasks, the project aims to quantify the benefits of domain adaptation, examine generalization trade-offs, and provide practical insight into when supervised fine-tuning meaningfully improves the performance of open language models for specialized NLP applications.

1 Introduction

Pretrained language models have become a central component of modern natural language processing systems, achieving strong performance across a wide range of general-purpose tasks. These models are typically trained on large, diverse text corpora and are designed to capture broad linguistic patterns. However, their effectiveness often diminishes when applied to specialized domains such as legal and medical text, where precise terminology, structured documents, and domain-specific reasoning are critical.

Domain-specific fine-tuning has emerged as a practical approach for adapting pretrained language models to specialized domains. By continuing training on domain-relevant datasets, fine-tuning allows models to better align their internal representations with specialized vocabulary and contextual patterns. Despite its widespread adoption, the benefits of fine-tuning are not always systematically evaluated across different domains and tasks, particularly for open-source language models operating under realistic computational constraints.

This project empirically evaluates the impact of supervised domain-specific fine-tuning on an open-source language model using legal and medical datasets. Model performance is assessed before and after fine-tuning on question answering and text generation tasks. Through controlled evaluation and quantitative metrics, the study aims to quantify performance improvements, assess domain specialization effects, and provide practical insights into the effective use of fine-tuned language models for specialized natural language processing applications.

Problem statement

Pretrained open-source language models have demonstrated strong general-purpose capabilities, yet their performance often degrades when applied to specialized domains such as legal and medical text. These domains exhibit distinct linguistic characteristics, including specialized terminology, complex document structures, and domain-specific reasoning patterns, which are not fully captured by general-purpose training data. As a result, models deployed without adaptation may produce inaccurate or unreliable outputs.

Supervised domain-specific fine-tuning is commonly used to align pretrained language models with specialized domains. While fine-tuning is widely adopted in practice, its impact on downstream task performance is not always systematically evaluated, particularly for open-source models under realistic compute constraints. In addition, fine-tuning may introduce trade-offs related to over-specialization, generalization, and training efficiency that are not always explicitly measured.

This project investigates the effectiveness of supervised fine-tuning for domain adaptation by empirically comparing model performance before and after fine-tuning on legal and medical datasets. Using question answering and text generation as evaluation tasks, the study aims to quantify performance gains, assess cross-domain generalization, and provide practical insight into when and how fine-tuning improves the effectiveness of open language models in specialized domains.

2 Hypothesis

Pretrained general-purpose language models are optimized for broad linguistic coverage but are not inherently designed to perform well on specialized domains such as legal and medical text. These domains require precise terminology, structured reasoning, and adherence to domain-specific conventions that are often underrepresented in general training corpora. As a result, general-purpose models are expected to exhibit lower performance on domain-specific question answering and text generation tasks when used without adaptation.

This project hypothesizes that applying supervised domain-specific fine-tuning techniques will significantly improve model performance on specialized tasks. In particular, instruction-based supervised fine-tuning using parameter-efficient approaches such as Low-Rank Adaptation (LoRA) is expected to enhance task alignment by allowing the model to adjust its representations based on domain-relevant examples. These techniques enable targeted adaptation while minimizing computational cost, making them suitable for practical deployment on small open-source models.

Furthermore, the project hypothesizes that effective fine-tuning can substantially reduce the performance gap between general-purpose pretrained models and domain-specific task requirements. By aligning the model's learned representations with legal and medical data distributions, fine-tuned models are expected to demonstrate higher accuracy in question answering and improved quality in text generation compared to their baseline counterparts. The experiments are designed to empirically validate these hypotheses through controlled before-and-after performance comparisons.

Proposed Models

- Phi
- LLaMA
- Qwen
- Mistral

3 Transformer Architecture

The selected models (Phi-3-Mini, LLaMA-3.2, and Qwen-2.5) follow a decoder-only Transformer architecture. Unlike encoder-decoder architectures such as T5 or BART, which separate input encoding and output generation into distinct components, decoder-only models integrate representation learning and autoregressive text generation within a single stack of masked self-attention layers.

In this study, domain adaptation is applied exclusively to decoder-only architectures. Parameter-efficient fine-tuning methods are implemented within the Transformer decoder blocks. For matrix-based approaches such as LoRA and AdaLoRA, low-rank adaptation modules are inserted into the self-attention projection matrices (e.g., query and value projections). For vector-based methods such as IA^3, learnable scaling vectors are applied to selected internal activations. In all configurations, the pretrained base model parameters remain frozen during fine-tuning.

4 Related Works

The Transformer architecture introduced by Vaswani et al. revolutionized sequence modeling by relying solely on self-attention mechanisms. Subsequent research has demonstrated that domain-specific adaptation of Transformer-based models significantly improves performance in specialized fields. In the medical domain, models such as BioBERT, ClinicalBERT, and PubMedBERT show that fine-tuning on biomedical corpora enhances question answering and inference tasks. Similarly, in the legal domain, Legal-BERT and CaseLaw-BERT adapt Transformer representations to highly structured legal text, yielding improved results on legal NLP benchmarks. These works collectively motivate a systematic empirical evaluation of supervised fine-tuning strategies across multiple domains and tasks, which this project aims to address.

Attention Is All You Need
Advances in Neural Information Processing Systems (NeurIPS)
Foundational paper introducing the Transformer architecture.[10]

BioBERT: a pre-trained biomedical language representation model for biomedical text mining
Bioinformatics
Seminal work demonstrating domain adaptation of Transformers for medical text.[8]

Domain-Specific Language Model Pretraining for Biomedical NLP
Proceedings of the ACL
Shows benefits of domain-specific training for biomedical NLP tasks[4]

LEGAL-BERT: The Muppets straight out of Law School
Proceedings of EMNLP
Canonical Transformer-based model for legal domain adaptation.[2]

Don't Stop Pretraining: Adapt Language Models to Domains and Tasks
Proceedings of ACL
Provides strong empirical evidence for continued pretraining and fine-tuning.[5]

PubMedQA: A Dataset for Biomedical Research Question Answering
Proceedings of EMNLP-IJCNLP
Widely used benchmark for evaluating medical question answering.[7]

5 Proposed Solution

The proposed solution adopts a controlled experimental approach to evaluate the effectiveness of domain-specific supervised fine-tuning for open-source language models. A pretrained open model is selected as the baseline and evaluated on legal and medical datasets to establish initial performance on question answering and text generation tasks. This baseline assessment provides a reference point for measuring the impact of domain adaptation.

The model is then fine-tuned separately on domain-specific corpora for the legal and medical domains using supervised fine-tuning techniques. Fine-tuning is performed under consistent architectural and training configurations to isolate the effect of domain-specific data. This process allows the model to adapt its internal representations to specialized vocabulary, structure, and reasoning patterns present in each domain.

Performance after fine-tuning is evaluated using standardized quantitative metrics and compared directly against baseline results. By analyzing before-and-after performance across tasks and domains, the proposed solution quantifies the benefits of fine-tuning while also examining trade-offs related to generalization and training cost. The results provide practical insight into the effectiveness of domain adaptation for open language models in specialized applications.

6 Proposed Experiments

This project adopts a supervised fine-tuning strategy centered on instruction-based learning combined with parameter-efficient fine-tuning (PEFT) techniques. Specifically, Low-Rank Adaptation (LoRA) adapters are employed to fine-tune open-source Transformer-based language models for domain-specific tasks in the legal and medical domains. Instruction-based fine-tuning reformulates training data into instruction–response pairs, enabling the model to learn how to generate task-appropriate outputs rather than merely predicting the next token. This approach is well-suited for both question answering and text generation tasks, while LoRA ensures that the computational overhead remains manageable by updating only a small subset of model parameters within the attention layers.

The experimental design begins with a baseline evaluation to establish the performance of pretrained models prior to any domain adaptation. In this experiment, models are evaluated out-of-the-box on legal and medical question answering datasets, as well as on a fixed set of domain-specific text generation prompts. Performance is measured using standard metrics such as accuracy and F1 score for question answering, along with automated or lightweight human evaluation for text generation quality. This baseline experiment provides a critical reference point for quantifying the impact of domain-specific fine-tuning and highlights inherent limitations of general-purpose pretrained models when applied to specialized domains.

Subsequent experiments focus on domain-specific fine-tuning and in-domain evaluation. Separate fine-tuning runs are conducted for the legal and medical domains using instruction-based supervised learning with LoRA adapters. After fine-tuning, each model is evaluated on corresponding in-domain tasks to measure performance gains relative to the baseline. Legal-domain experiments assess improvements in legal question answering accuracy and legal text generation quality, while medical-domain experiments mirror this setup for medical question answering and generation tasks. Together, these experiments enable a controlled empirical comparison of how supervised fine-tuning reshapes model behavior across domains, providing insights into domain specialization, performance improvements, and trade-offs associated with efficient adaptation techniques.

Instruction-Based Supervised Fine-Tuning
Objective:
To enable the language model to learn task-oriented response behavior by explicitly training it to follow instructions and generate appropriate answers for domain-specific question answering and text generation tasks.

Method:

Training data is reformatted into an instruction–response structure, where each example consists of an instruction or question paired with a corresponding target answer. This format encourages the model to learn how to respond to prompts rather than simply predicting the next token in a sequence. Instruction-based supervised fine-tuning is applied consistently across legal and medical datasets to support both question answering and text generation tasks.

Purpose:
This approach improves task alignment by teaching the model how to generate structured, context-aware responses that are appropriate for domain-specific applications. Instruction-based fine-tuning has been shown to be particularly effective for improving performance on interactive tasks such as question answering and controlled text generation.

Parameter-Efficient Fine-Tuning Using LoRA
Objective:
To adapt pretrained language models to specialized domains while minimizing computational cost and reducing the risk of overfitting.

Method:
Parameter-efficient fine-tuning (PEFT) is implemented using Low-Rank Adaptation (LoRA). Instead of updating all model parameters, LoRA injects low-rank adapter matrices into selected attention layers of the Transformer architecture. During training, only these adapter parameters are updated, while the original model weights remain frozen.

Purpose:
This technique enables efficient domain adaptation by significantly reducing memory usage and training time, making fine-tuning feasible for small open-source models and limited computational resources. LoRA is widely used and well-validated in academic research, providing a robust and reproducible mechanism for domain-specific supervised fine-tuning.

6.1 Baseline Evaluation (Before Fine-Tuning)

Objective:
To understand how pretrained open-source language models perform on legal and medical tasks before any domain-specific training is applied.

Method:
Each selected model is evaluated in its original, pretrained form on the following tasks:

- Legal question answering datasets
- Medical question answering datasets
- A fixed set of domain-specific text generation prompts

No additional training or adaptation is performed during this stage.

Evaluation Metrics:

- Question answering accuracy and F1 score
- Text generation quality, measured using automatic similarity-based metrics and supplemented with limited human judgment for correctness and clarity

Purpose:
This experiment establishes a baseline reference point and highlights inherent differences in domain knowledge and reasoning ability across models prior to fine-tuning. These results are used for comparison in all subsequent experiments.

6.2 Legal Domain Fine-Tuning and In-Domain Evaluation

Objective:
To evaluate the effect of legal domain fine-tuning on legal question answering and text generation performance.

Method:
Each model is fine-tuned on legal domain training data using instruction-based supervised fine-tuning with LoRA adapters. The adapted models are then evaluated on legal QA datasets and legal text generation prompts that were not seen during training.

Evaluation Metrics:

- Legal QA accuracy and F1 score
- Legal text generation quality

Purpose:
This experiment measures the extent to which legal domain fine-tuning improves in-domain performance relative to the baseline established in Experiment 1.

6.3 Medical Domain Fine-Tuning and In-Domain Evaluation

Objective:
To evaluate the effect of medical domain fine-tuning on medical question answering and text generation performance.

Method:
Each model is fine-tuned on medical domain training data using the same instruction-based fine-tuning approach. The resulting models are evaluated on medical QA datasets and medical text generation prompts.

Evaluation Metrics:

- Medical QA accuracy and F1 score
- Medical text generation quality

Purpose:
This experiment mirrors Experiment 2 for the medical domain, enabling a direct comparison of how domain-specific adaptation affects performance across different specialized domains.

6.4 Cross-Domain Generalization Analysis

Objective:
To examine whether fine-tuning on one domain improves specialization at the cost of performance in another domain.

Method:

- Models fine-tuned on legal data are evaluated on medical QA datasets
- Models fine-tuned on medical data are evaluated on legal QA datasets

The results are compared against baseline performance from Experiment 1.

Evaluation Metrics:
Cross-domain QA accuracy and F1 score relative to baseline

Purpose:
This experiment investigates potential specialization trade-offs and domain interference introduced by fine-tuning, providing insight into how domain adaptation affects generalization.

6.5 Training Data Size Ablation

Objective:
To analyze how model performance changes as the amount of domain-specific training data increases.

Method:
A selected model is fine-tuned using different proportions of domain-specific training data (for example, 10%, 50%, and 100%) for a single domain. In-domain performance is evaluated after each training configuration.

Evaluation Metrics:
QA accuracy and F1 score plotted against training data size

Purpose:
This experiment examines learning efficiency and identifies whether performance gains exhibit diminishing returns as more training data is added.

6.6 LoRA versus Full Fine-Tuning Comparison

Objective:
To compare parameter-efficient fine-tuning with full fine-tuning in terms of performance improvement and computational cost.

Method:
For a selected model and domain, two fine-tuning configurations are evaluated:

- LoRA-based parameter-efficient fine-tuning
- Full fine-tuning of all model parameters

Both configurations use identical training data, evaluation datasets, and training schedules.

Evaluation Metrics:

- QA accuracy and F1 score
- Text generation quality
- Training time and GPU memory usage

Purpose:
This experiment provides a practical comparison between efficiency and performance, helping determine whether parameter-efficient methods can achieve results comparable to full fine-tuning at lower computational cost.

6.7 Error and Hallucination Analysis

Objective:
To analyze how fine-tuning affects error patterns and hallucination behavior in generated text.

Method:
A representative subset of incorrect or low-quality outputs (approximately 50–100 samples) is manually reviewed before and after fine-tuning.

Evaluation Categories:

- Fabricated or unsupported facts
- Incorrect domain-specific details

- Vague or incomplete responses
- Overconfident but incorrect assertions

Purpose:
This analysis complements quantitative metrics by identifying failure modes that are not fully captured by automatic evaluation scores. It also helps explain differences in hallucination behavior across models and training stages.

6.8 Dynamic Rank Allocation with AdaLoRA

Objective:
Evaluate whether dynamic rank allocation (AdaLoRA) yields better domain adaptation performance than fixed-rank LoRA under a comparable parameter budget.

Method:
Compare two PEFT strategies using the same base model, dataset splits, training steps, and optimizer settings.

- Control (LoRA): Apply a fixed rank (e.g., $r = 16$) uniformly to the same target modules across all decoder blocks (e.g., self-attention q_proj and v_proj).
- Experimental (AdaLoRA): Initialize with a rank budget and allow AdaLoRA to dynamically reallocate rank across layers during training based on learned importance, assigning higher effective rank to more impactful layers and reducing rank in less critical layers.

Evaluation Metrics:

- Task performance: QA Accuracy / F1; generation metrics (ROUGE/BLEU where applicable)
- Parameter efficiency: Performance vs. number of trainable parameters
- Layer-wise allocation analysis: Rank distribution across layers (early vs. late; attention vs. other modules, if applicable)

Purpose:
Test the hypothesis that different layers contribute unequally to domain adaptation, and that adaptive allocation can improve performance given a limited adapter budget.

6.9 Extreme Parameter Efficiency with IA^3

Objective:
Evaluate whether vector-based activation rescaling (IA^3) can achieve comparable domain adaptation performance to LoRA with substantially fewer trainable parameters.

Method:
Implement IA^3 by introducing learnable scaling vectors that modulate internal activations via element-wise multiplication within Transformer blocks (e.g., scaling attention key/value pathways and FFN intermediate activations). Compare IA^3 against the LoRA control using the same training protocol.

Evaluation Metrics:

- Task performance: QA Accuracy / F1; generation quality metrics
- Parameter density: Trainable parameter count vs. performance
- Storage overhead: Saved adapter size on disk
- Compute/memory: Peak GPU memory usage; training throughput
- Inference latency (measured): Report any observed differences (not assumed)

Purpose:
Probe the "minimum adaptation" needed for domain transfer—whether adaptation requires weight-space updates (LoRA) or can be achieved by reweighting existing features (IA^3).

6.10 Quantization Effects and Resource Efficiency (Standard LoRA vs. QLoRA)

Objective:
To examine the trade-off between computational resource efficiency and downstream task performance when applying 4-bit quantization (QLoRA) compared to standard 16-bit LoRA fine-tuning.

Method:
We will fine-tune a large language model on the selected domain-specific corpus using two configurations:

- Baseline (Standard LoRA): The base model is loaded in 16-bit precision (FP16 or BF16), and LoRA adapters are trained under standard precision settings.
- Experimental (QLoRA): The base model is loaded in 4-bit precision using the NormalFloat4 (NF4) data type with double quantization. LoRA adapters are trained in 16-bit precision while the quantized base model weights remain frozen.

During training, we will record peak GPU memory usage and measure the final adapter size.

Evaluation Metrics:

- Resource Metrics: Peak GPU memory usage (GB); Training time per epoch
- Performance Metrics: Accuracy and F1 score on the domain validation set to assess potential degradation due to quantization

Purpose:
This experiment evaluates whether QLoRA enables efficient domain adaptation of large language models on limited hardware without significant performance degradation. The results will help determine whether 4-bit quantization provides a practical pathway for deploying domain-adapted models on consumer-grade GPUs while maintaining competitive task accuracy.

7 Experiment Set-Up and Data-Set Details

This project uses open-source, domain-specific datasets to evaluate the effects of supervised fine-tuning on open language models in the legal and medical domains. All datasets are publicly available, well-documented, and commonly used in academic research, making them suitable for systematic evaluation and reproducibility. The selected datasets include both long-form domain text and structured question–answer pairs, allowing the study to assess model performance across different task types.

To ensure consistency across experiments, datasets are preprocessed using standardized formatting and prompt structures. Training and evaluation splits provided by the datasets are used wherever available to avoid data leakage. All datasets are used strictly for research purposes.

Legal Domain Datasets:

The primary legal corpus is obtained from the Caselaw Access Project (CAP), maintained by Harvard Law School's Library Innovation Lab. This dataset contains digitized U.S. court opinions spanning multiple jurisdictions and time periods. The corpus provides long-form, highly structured legal text, making it well suited for domain adaptation through fine-tuning.

To support evaluation on legal question answering tasks, the project additionally leverages LegalQAEval, an open dataset hosted on Hugging Face. This dataset provides question–answer pairs grounded in legal content, enabling standardized evaluation of model performance before and after fine-tuning.

Legal dataset sources:

- Caselaw Access Project: https://case.law/

- LegalQAEval (Hugging Face): https://huggingface.co/datasets/isaacus/LegalQAEval

Medical Domain Datasets:
For the medical domain, the project primarily uses PubMedQA, accessed via the BigBio collection on Hugging Face. PubMedQA consists of biomedical research questions paired with supporting PubMed abstracts and labeled answers. This dataset is widely used for evaluating medical question answering systems and provides a reliable benchmark for domain-specific adaptation.
In addition, MedQuAD (Medical Question Answering Dataset) is used to supplement the medical evaluation. MedQuAD contains curated medical question–answer pairs sourced from trusted medical websites, offering diverse coverage of clinical and biomedical topics.

Medical dataset sources:

- PubMedQA (BigBio): https://huggingface.co/datasets/bigbio/pubmed_qa

- MedQuAD (Hugging Face): https://huggingface.co/datasets/keivalya/MedQuad-MedicalQnADataset

8 Proposed Evaluation Metrics

The evaluation framework for this project is designed to quantitatively measure the impact of domain-specific supervised fine-tuning on open-source language models. Model performance is evaluated both before and after fine-tuning to enable controlled comparison. Metrics are selected to reflect the requirements of legal and medical domains, where correctness, relevance, and clarity are critical. The evaluation focuses on two primary task categories: question answering and text generation.

For question answering (QA) tasks in both legal and medical domains, performance is measured using Accuracy and F1 score. Accuracy captures the proportion of correctly answered questions and is suitable for classification-style QA tasks. The F1 score is used to account for partial correctness in cases where answers may vary in phrasing or contain overlapping information. These metrics together provide a balanced assessment of model understanding and response quality across domains.

For text generation tasks, automatic evaluation metrics such as ROUGE and BLEU are employed when reference outputs are available. ROUGE measures n-gram overlap and is commonly used for summarization and explanation tasks, while BLEU provides precision-based evaluation for structured generation. These metrics offer a standardized way to compare generation quality before and after fine-tuning and to quantify improvements in domain-specific language generation.

8.1 Evaluation Metrics Summary

Task	Metric
Legal QA	Accuracy F1
Medical QA	Accuracy F1
Text Generation	ROUGE, BLEU

8.2 Metric Justification

Accuracy is used as a primary evaluation metric for question answering tasks in both the legal and medical domains. It provides a straightforward measure of the proportion of correctly answered questions and is particularly suitable for classification-style or short-answer QA tasks. Accuracy

enables clear before-and-after comparison and allows direct measurement of performance gains resulting from domain-specific fine-tuning.

F1 score complements accuracy by capturing partial correctness in cases where predicted answers may not exactly match ground-truth responses but still contain relevant information. This is especially important for legal and medical question answering, where responses may vary in phrasing while remaining semantically correct. The F1 score provides a more nuanced evaluation of model performance by balancing precision and recall.

ROUGE is employed to evaluate text generation quality by measuring n-gram overlap between generated outputs and reference texts. It is widely used in summarization and explanation tasks and serves as an effective indicator of content relevance and coverage. ROUGE allows systematic comparison of generation quality before and after fine-tuning in domain-specific contexts.

BLEU is optionally used as a supplementary metric for text generation tasks that involve more structured outputs. BLEU focuses on precision-based n-gram matching and is useful when evaluating shorter or more formulaic generated responses. While BLEU alone may not fully capture semantic quality, it provides additional quantitative insight when used alongside ROUGE.

9 Project Execution Timeline

- Detailed Project Execution Timeline (Jan 28 – Mar 17)
- Phase 1: Project Setup (Jan 28 – Feb 2)
- Phase 2: Dataset Acquisition & Preprocessing (Feb 3 – Feb 9)
- Phase 3: Baseline Model Evaluation (Feb 10 – Feb 14)
- Phase 4: Domain-Specific Fine-Tuning (Feb 15 – Feb 25)
- Phase 5: Post-Fine-Tuning Evaluation & Cross-Domain Testing (Feb 26 – Mar 5)
- Phase 6: Analysis & Interpretation (Mar 6 – Mar 10)
- Phase 7: Report Writing & Presentation Preparation (Mar 11 – Mar 15)
- Phase 8: Final Review & Submission (Mar 16 – Mar 17)

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